

Education

University of Bath - 2019-2024

MSc Computer Science
Grade: Distinction

BEng (Hons) Civil Engineering
Grade: First Class Honours

Stewart's Melville College - 2006-2019

Advanced Highers (A*A*A)

Highers (A*A*A*A*A)

National 5 (6A* and 2A)

Work Experience

IMP Software

Sep 2024 – Present

Junior Software Developer

- Contributed to the Purchasing Squad, a SCRUM team of six developers and two testers, responsible for delivering a new end-to-end purchasing system tailored for Multi Academy Trusts.
- Developed a Blazor web application integrated with Azure Function Apps, enabling seamless cloud-based workflows and enhancing system scalability and performance.
- Played an integral role in application development by completing user stories, troubleshooting and resolving bugs, participating in code reviews, refining tickets, and deploying code in a fast-paced Agile environment.
- Gained hands-on experience with the full software lifecycle, from conceptualisation to deployment, contributing directly to the successful launch of the product.
- Utilised a technology stack including C#, Blazor, .NET, SQL, and Azure DevOps to build and maintain a robust and efficient solution.

System C Healthcare

Jun 2023-Sep 2023

Software Engineer Intern

- As part of the Systems Interoperability team, contributed to a SCRUM group focused on resolving critical bugs and improving the client interface for enhanced user experience.
- Worked alongside the R&D team to design and develop a Blazor web application aimed at improving efficiency for the customer support team.
- Led the development of both the front-end and back-end of the web app, ensuring seamless integration with a web service built by the lead engineer.
- Ensured that the web application met all functional requirements and adhered to the technical specifications set out by the lead engineer.

Rendel Ltd

Aug 2021-Jul 2022

Undergraduate Civil Engineer

- Worked as part of the bridges team, which focussed on the category three checks of the viaducts that are being built as part of the HS2 Railway Project.
 - This role involved creating 3D models of the bridges for Finite Element Analysis, using Excel spreadsheets to perform calculations on structural members and writing Python scripts that would help to input thousands of rows of data into modelling software.
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Interests

My main two interests are cooking and rugby. I started playing rugby at primary school and continued to do so whilst at university. Some of my highlights include being in the First XV at High School, being selected for Edinburgh U19's and U20's, playing for the University of Bath's first team in BUCS Super Rugby and playing for Scotland at the Junior Touch World Championship. Playing rugby has improved my teamwork skills, commitment and determination. I am passionate about food and cooking and spend my free time reading about and experimenting with different recipes and cuisines.

Technical Skills

Languages:

- **C#** - I am very experienced with C# and have proven this by developing multiple Blazor web applications, both in industry and for personal projects.
- **Python** - I have been writing Python code since 2019 and am extremely proficient in all aspects of the language. I have modelled complex structural engineering calculations and built AI search algorithms and Machine Learning models.
- **Swift** - I taught myself Swift to be able to build an iOS app for the Apple Vision Pro. I am proficient with the fundamentals of the language.
- **C++** - I have used C++ to program Arduinos and other microcontrollers as part of my undergraduate dissertation project.

Frameworks:

- **Blazor** - I taught myself the core concepts of Blazor using Pluralsight courses and have since gone on to build multiple web applications as both personal projects and through working as a software engineer.
- **iOS** - As part of my university dissertation I designed and built an iOS app for the Apple Vision Pro.
- **Django** - Using William S. Vincent's book 'Django for Professionals' I taught myself how to use Django as a framework to create web applications.

Other:

- **Data Science** - Three weeks completing the Codecademy 'Data Science Career Path' course which included Python, SQL, NumPy, Pandas, data visualisation, natural language processing, and machine learning. This valuable course taught me the basics of data science and analytics.
- **Artificial Intelligence** - During my master's degree in computer science I learned to build AI search algorithms and Machine Learning models. Including MDPs, Monte Carlo, Temporal Difference, Function Approximation, Depth and Breadth First, Supervised Learning, and Neural Networks.

Projects – More detail on these projects can be found on [my website](#).

Enhancing Neonatal Intensive Care Through Mixed Reality Patient Monitoring

Explored the potential application of mixed reality (MR) in neonatal maternity wards to enhance patient monitoring. The project focused on developing a proof-of-concept MR application that integrates real-time physiological data from cot-side monitors and patient care records. Key benefits identified include improved patient privacy, hands-free data visualization for better hygiene practices, and the ability to view dynamic and spatial data. Despite the promising advantages, MR's use in neonatal healthcare remains limited, which this research aimed to address. The project involved selecting appropriate technology, conducting design workshops with clinical specialists, and developing two full-stack MR applications. The applications were evaluated through user feedback to assess their impact on patient monitoring and clinical outcomes.

Development of a Low-Cost Water Level Gauge

Conducted a research project to investigate the feasibility of low-cost water level gauges and compare their accuracy with existing solutions. Developed two different gauges using an ESP32-CAM, Arduino, and ultrasonic sensor, programmed with Arduino IDE and C++. The ESP32-CAM captured images of the water level at regular intervals, which were processed using custom Python software I developed. This software employed edge-detecting algorithms and a reference distance to calculate the water depth from the images. Additionally, an Arduino Nano and ultrasonic sensor were used to measure distance and track changes in water depth. Both gauges were designed to operate off-grid with battery power and stored data on micro-SD cards. The research concluded that both gauges could serve as effective, low-cost alternatives to traditional water level monitoring solutions.